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United States Patent	12410663
Kind Code	B2
Date of Patent	September 09, 2025
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Completion string, method, and system

Abstract

A completion string, including an upper completion having a shifting tool, and a lower completion having a hydraulic releaser and a valve responsive to the shifting tool. A method for completing a well, including running the completion string in a borehole, establishing pressure integrity, applying tubing pressure to release the hydraulic releaser, and shifting the valve. A method for completing a well, including running an upper completion connected to a lower completion into a borehole, releasing a hydraulic releaser with applied tubing pressure, and shifting a valve in the lower completion with movement of the upper completion. A wellbore system, including a borehole in a subsurface formation, an upper completion and a lower completion in the borehole, a hydraulic releaser connecting the upper completion to the lower completion, the hydraulic releaser being responsive to applied tubing pressure to releaser the upper completion from the lower completion.

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Appl. No.: 18/463871

Filed: September 08, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20250084710 A1	Mar. 13, 2025

Publication Classification

Int. Cl.: E21B17/06 (20060101); E21B23/04 (20060101); E21B34/12 (20060101); E21B43/10 (20060101)

U.S. Cl.:

CPC E21B17/06 (20130101); E21B23/04 (20130101); E21B34/12 (20130101); E21B43/10 (20130101);

Field of Classification Search

CPC: E21B (34/12); E21B (43/10); E21B (23/04); E21B (17/06)

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Background/Summary

BACKGROUND

(1) In the resource recovery and fluid sequestration industries there is often need for what are known as lower completions and upper completions. These are generally run separately and manipulated using shear up or shear down processes to render the completions fit to convey fluid or be separated for other operations. Shear up/shear down works reasonably well but leaves some efficiencies off the table. The art is always receptive to efficiency increasing technologies.

SUMMARY

- (2) An embodiment of a completion string, including an upper completion having a shifting tool, and a lower completion having a hydraulic releaser and a valve responsive to the shifting tool.
- (3) An embodiment of a method for completing a well, including running the completion string in a

borehole, establishing pressure integrity, applying tubing pressure to release the hydraulic releaser, and shifting the valve.

(4) An embodiment of a method for completing a well, including running an upper completion connected to a lower completion into a borehole, releasing a hydraulic releaser with applied tubing pressure, and shifting a valve in the lower completion with movement of the upper completion.

(5) An embodiment of a wellbore system, including a borehole in a subsurface formation, an upper completion and a lower completion in the borehole, a hydraulic releaser connecting the upper completion to the lower completion, the hydraulic releaser being responsive to applied tubing pressure to releaser the upper completion from the lower completion.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The following descriptions should not be considered limiting in any way. With reference to the accompanying drawings, like elements are numbered alike:

(2) FIG. 1 is a sectional view of a completion as disclosed herein;

(3) FIGS. 2-8 are views of the same completion in various operational positions;

(4) FIGS. 9-11 are views of the hydraulic releaser as disclosed herein in various operational positions;

(5) FIG. 12 is a sectional view of a hydraulic releaser profile;

(6) FIG. 13 is a side view of the hydraulic releaser profile;

(7) FIG. 14 is a view of a wellbore system including the completion as disclosed herein.

DETAILED DESCRIPTION

(8) A detailed description of one or more embodiments of the disclosed apparatus and method are presented herein by way of exemplification and not limitation with reference to the Figures.

(9) FIG. 1 is an overview of the completion 10 as disclosed herein that provides a frame of reference for the disclosure. Completion 10 includes a lower completion 12 and an upper completion 14. The upper completion nests radially inwardly with the lower completion 12 when the two are connected. The lower completion may include, in embodiments, an entry guide 16, a seal bore 18, a hydraulic release module 20, a tubing space out joint (TSOJ) housing 22, a lower completion tubing valve assembly 24, (which itself includes a one-way profile 26, a valve profile 28 distinct from the one-way profile 26, and a valve 30), a lower completion packer/anchor 32, a lower completion tubing 34, a lower completion radial valve 36, a screen 38, a toe valve 40 (well isolation valve), and a float shoe 42. The upper completion 14 may include, in embodiments, a tubing hanger 44, a safety valve 46, an upper completion tubing 48, an upper completion packer 50, a circulating valve 52, a hydraulic release sub 54 (which is interactive with the hydraulic release module 20 and together therewith forms a hydraulic releaser 56), a seal mandrel 58, a number of seals 60, a washpipe 62, and a one-way shifter profile 64. Each of the components listed for the lower completion 12 and upper completion 14 are connected as illustrated in FIG. 1.

(10) The completion 10 is runnable to target depth in a connected form in one run and then may be manipulated to release the upper completion 14 from the lower completion 12 in order to stroke the upper completion. While that statement in itself is not new, taking this action via a hydraulic releaser 56 and allowing uphole direction and downhole direction movement of the upper completion relative to the lower completion is not found in the art. Traditionally, a shear up or shear down is needed where no such action is required in the completion 10 disclosed herein. Rather, efficiency is dramatically improved by configuring the completion 10 to respond to a hydraulic input to release the upper completion 14 from the lower completion 12. This configuration facilitates conveyance of the completion into a wellbore with high axial loads that might prematurely activate a prior art shear-type release mechanism. In this context, “high” means

upwards of about 50,000 pounds. The configuration also facilitates uphole direction and downhole direction movement options. This enables placing the tubing hanger **44** in a precise predetermined location, while placing the lower completion **12** in a variable location to react to varying conditions in the lower wellbore. Incorporating the shifter **64** in the upper completion facilitates efficient closing of the valve **30** with movement of the upper completion. At the conclusion of operations, the upper completion **14** is permanently installed and not removed from the well since with the configuration as disclosed does not employ a temporary workstring (that would have to be removed from the well) like the prior art does thereby enabling greater efficiency of the present disclosure.

(11) Referring to FIGS. **2-8**, a sequence of operations is illustrated with the lower completion generally maintained in position so that moving components are easy to appreciate. It will be appreciated that FIG. **2** is very similar to FIG. **1**. This is the way that completion **10** is run in the hole. Once the completion is at target depth in a borehole **92** (see FIG. **14**), an object **66** (FIG. **3**) is released into the completion **10** to land on the toe valve or well isolation valve seat **40**. Pressure integrity is established and pressure applied to tubing or to annulus may be used to set the lower completion packer **32**. It is the differential in tubing and annulus pressure that causes the set sequence in ways that are known in the art. Once the packer **32** is set (FIG. **4**), the pressure differential is increased to activate the hydraulic releaser **56**, which eliminates the axial and rotational lock between the lower completion **12** and the upper completion **14** (FIG. **5**). Once the releaser **56** releases, it is possible to pick up on the upper completion **14** leaving the lower completion **12** in place (and anchored by the packer **32**). Picking up on the upper completion **14** (FIG. **6**) allows the shifter profile **64** to shift the valve **30** to a closed position by engaging and moving profile **26**. It is to be understood that shifter profile **64** only engages and moves the profile **26** in a single direction, that direction associated with the closing of the valve **30**. Shifter profile **64** interacting with the profile **26** in the valve open direction will simply bump past and not engage. The upper completion may then be set down with slack off weight to land the tubing hanger **44** (FIG. **7**). Finally, tubing pressure is applied to set the upper completion packer **50**.

(12) Referring to FIGS. **9-11**, an enlarged view of the hydraulic releaser **56** is illustrated. It will be appreciated that a dog **68** extends through the tubing space out joint housing **22** into contact with the hydraulic release sub **54**. As such, the upper completion **14** and lower completion **12** are axially and rotationally affixed to one another. The dog **68** is maintained in this position by a releaser housing **70** that is sealed to the TSOJ housing **22** but movable thereon in response to applied pressures. Seals **72** are illustrated having different diameters to respond to differential pressures from either the tubing to annulus or annulus to tubing. The housing **70** includes recesses **74** on either side of a holder **76** such that movement of the housing **70** in either longitudinal direction of the completion **10** will result in the dog **68** being unlocked and a biaser **78** moving the dog **68** radially outwardly into either recess **74**. As noted, the movement of the housing **70** is due to pressure, that pressure being applied through port **80** in TSOJ housing **22**. In some embodiments, the releaser **56** will include a shear member **82** or equivalent to hold the housing **70** in position until a threshold pressure differential is attained across seals **72**. In the released position, illustrated in FIGS. **10** and **11** (in differing longitudinal directions) it can be seen that the dog **68** and the hydraulic release sub **54** are no longer engaged, hence allowing freedom of movement between upper completion **14** and lower completion **12**.

(13) Referring to FIGS. **12** and **13**, an enlarged view of the hydraulic release sub **54** is illustrated. The views illustrate castellations **84** for rotational locking and castellations **86** for axial locking.

(14) Referring to FIG. **14**, a borehole system **90** is illustrated. The system **90** comprises a borehole **92** in a subsurface formation **94**. A completion **10** as disclosed herein is disposed within the borehole **92**.

(15) Set forth below are some embodiments of the foregoing disclosure:

(16) Embodiment 1: A completion string, including an upper completion having a shifting tool, and a lower completion having a hydraulic releaser and a valve responsive to the shifting tool.

(17) Embodiment 2: The string as in any prior embodiment, wherein the hydraulic releaser is downhole of a seal bore in the lower completion.

(18) Embodiment 3: The string as in any prior embodiment, wherein the hydraulic releaser locks the upper and lower completions together until released.

(19) Embodiment 4: The string as in any prior embodiment, wherein upon release of the hydraulic release the upper completion is movable both in the uphole direction and downhole direction from the position where the upper completion was locked by the hydraulic releaser.

(20) Embodiment 5: The string as in any prior embodiment, wherein the hydraulic releaser includes features that axially and rotationally lock the upper and lower completions relative to one another.

(21) Embodiment 6: The string as in any prior embodiment, wherein the lower completion further includes a wellbore isolation valve.

(22) Embodiment 7: The string as in any prior embodiment, wherein the hydraulic releaser is responsive to differential pressure inside the completion relative to pressure outside of the completion.

(23) Embodiment 8: The string as in any prior embodiment, wherein the upper completion is configured to be permanently installed in a wellbore.

(24) Embodiment 9: The string as in any prior embodiment, further comprising a lower completion tubing valve assembly that includes a one-way profile to open a valve of the assembly and a distinct valve profile to close the valve.

(25) Embodiment 10: A method for completing a well, including running the completion string as in any prior embodiment in a borehole, establishing pressure integrity, applying tubing pressure to release the hydraulic releaser, and shifting the valve.

(26) Embodiment 11: The method as in any prior embodiment, wherein the establishing pressure integrity includes dropping an object and seating the object in a seat in the lower completion.

(27) Embodiment 12: The method as in any prior embodiment, wherein the applying pressure further includes setting a lower completion anchor.

(28) Embodiment 13: The method as in any prior embodiment, wherein the shifting includes picking up on the upper completion.

(29) Embodiment 14: The method as in any prior embodiment, further comprising landing a tubing hanger.

(30) Embodiment 15: The method as in any prior embodiment, further including setting an upper completion packer with applied tubing pressure.

(31) Embodiment 16: A method for completing a well, including running an upper completion connected to a lower completion into a borehole, releasing a hydraulic releaser with applied tubing pressure, and shifting a valve in the lower completion with movement of the upper completion.

(32) Embodiment 17: The method as in any prior embodiment, further including permanently installing the upper completion in the well.

(33) Embodiment 18: A wellbore system, including a borehole in a subsurface formation, an upper completion and a lower completion in the borehole, a hydraulic releaser connecting the upper completion to the lower completion, the hydraulic releaser being responsive to applied tubing pressure to releaser the upper completion from the lower completion.

(34) Embodiment 19: The system as in any prior embodiment, wherein the upper completion includes a shifting tool and the lower completion includes a valve responsive to the shifting tool.

(35) The use of the terms “a” and “an” and “the” and similar referents in the context of describing the invention (especially in the context of the following claims) are to be construed to cover both the singular and the plural, unless otherwise indicated herein or clearly contradicted by context. Further, it should be noted that the terms “first,” “second,” and the like herein do not denote any order, quantity, or importance, but rather are used to distinguish one element from another. The terms “about,” “substantially” and “generally” are intended to include the degree of error associated with measurement of the particular quantity based upon the equipment available at the

time of filing the application. For example, “about” and/or “substantially” and/or “generally” can include a range of +8% of a given value.

(36) The teachings of the present disclosure may be used in a variety of well operations. These operations may involve using one or more treatment agents to treat a formation, the fluids resident in a formation, a borehole, and/or equipment in the borehole, such as production tubing. The treatment agents may be in the form of liquids, gases, solids, semi-solids, and mixtures thereof. Illustrative treatment agents include, but are not limited to, fracturing fluids, acids, steam, water, brine, anti-corrosion agents, cement, permeability modifiers, drilling muds, emulsifiers, demulsifiers, tracers, flow improvers etc. Illustrative well operations include, but are not limited to, hydraulic fracturing, stimulation, tracer injection, cleaning, acidizing, steam injection, water flooding, cementing, etc.

(37) While the invention has been described with reference to an exemplary embodiment or embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the invention. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the invention without departing from the essential scope thereof. Therefore, it is intended that the invention not be limited to the particular embodiment disclosed as the best mode contemplated for carrying out this invention, but that the invention will include all embodiments falling within the scope of the claims. Also, in the drawings and the description, there have been disclosed exemplary embodiments of the invention and, although specific terms may have been employed, they are unless otherwise stated used in a generic and descriptive sense only and not for purposes of limitation, the scope of the invention therefore not being so limited.

Claims

1. A completion string, comprising: an upper completion having a shifting tool; and a lower completion having a hydraulic releaser and a valve responsive to the shifting tool, wherein the hydraulic releaser locks the upper and lower completions together until released.
2. The string as claimed in claim 1, wherein the hydraulic releaser is downhole of a seal bore in the lower completion.
3. The string as claimed in claim 1, wherein upon release of the hydraulic releaser the upper completion is movable both in the uphole direction and downhole direction from the position where the upper completion was locked by the hydraulic releaser.
4. The string as claimed in claim 1, wherein the hydraulic releaser includes features that axially and rotationally lock the upper and lower completions relative to one another.
5. The string as claimed in claim 1, wherein the lower completion further includes a wellbore isolation valve.
6. The string as claimed in claim 1, wherein the hydraulic releaser is responsive to differential pressure inside the lower completion relative to pressure outside of the lower completion.
7. The string as claimed in claim 1, further comprising a lower completion tubing valve assembly that includes a one-way profile to open the valve of the assembly and a distinct valve profile to close the valve.
8. A method for completing a well, comprising: running the completion string as claimed in claim 1 in a borehole; establishing pressure integrity; applying tubing pressure to release the hydraulic releaser; and shifting the valve.
9. The method as claimed in claim 8, wherein the establishing pressure integrity includes dropping an object and seating the object in a seat in the lower completion.
10. The method as claimed in claim 8, wherein the applying pressure further includes setting a lower completion anchor.
11. The method as claimed in claim 8, wherein the shifting includes picking up on the upper

completion.

12. The method as claimed in claim 8, further comprising landing a tubing hanger.

13. The method as claimed in claim 8, further including setting an upper completion packer with applied tubing pressure.

14. A method for completing a well, comprising: running an upper completion locked to a lower completion by a hydraulic releaser into a borehole; releasing the hydraulic releaser with applied tubing pressure; and shifting a valve in the lower completion with a shifting tool of the upper completion.

15. A wellbore system, comprising: a borehole in a subsurface formation; an upper completion having a shifting tool and a lower completion having a hydraulic releaser and a valve responsive to the shifting tool in the borehole; a hydraulic releaser locking the upper completion to the lower completion until released, the hydraulic releaser being responsive to applied tubing pressure to releases the upper completion from the lower completion.
